Myopic Sampling Can Be Maximally Suboptimal in Risk-Limiting Financial Audits

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August 11, 2026

Abstract

Risk-limiting financial audits estimate a weighted finite-population total by adaptive sampling without replacement. Shekhar, Xu, Lipton, Liang, and Ramdas derive a one-step oracle that samples transaction (i) proportionally to its monetary misstatement contribution $\pi_i f_i$, while leaving the multistage optimal-policy problem open. We show that repeating this oracle can have the worst possible finite-horizon approximation ratio. For every $N \geq 2$, at the fixed risk limit $\delta = 0.05$, an exact rational family has optimal expected audit length 1 and oracle expectation

$$1 + \frac{N-1}{1+\rho},$$

which approaches N as $\rho \downarrow 0$. The proof controls the betting, logical, and running-intersection components of the confidence sequence. Perfect point scores do not help: when $S = f$, prop-MS equals the oracle and inherits the lower bound.

We then give a positive, implementable result for certified AI uncertainty. If simultaneous intervals $\ell_i \leq f_i \leq u_i$ are available, auditing in descending order of $\pi_i(u_i - \ell_i)$ is pathwise minimax-optimal for the resulting robust certificate, for every finite population. We give the matching lower bound, an exact heterogeneous-cost dynamic program, a valid risk-budget composition with betting confidence sequences, exact finite-grid Bellman verification, and reproducible synthetic stress tests. The results distinguish next-draw variance reduction from the terminal value of entering an audit's stopping region.

1 Introduction

Let N transactions have normalized recorded-value weights $\pi_i > 0$, $\sum_i \pi_i = 1$, and unknown misstatement fractions $f_i \in [0, 1]$. The target is the total monetary misstatement

$$m^\star = \sum_{i=1}^{N} \pi_i f_i.$$

At each round an auditor chooses a predictable distribution q_t on the remaining transactions, observes one f_i, updates an anytime-valid confidence sequence, and stops when its diameter is at most ε. Shekhar et al. [1] introduced this risk-limiting financial audit (RLFA) framework. Their Proposition 2 minimizes the next importance-weighted observation's variance and maximizes a lower bound on one-step wealth growth using

$$q_t^{\text{oracle}}(i) = \frac{\pi_i f_i}{\sum_{j \in R_t} \pi_j f_j}. \tag{1}$$

They explicitly describe the actual fastest-reduction problem as a difficult multistage optimization and leave a fuller characterization for future work. The issue is consequential: the confidence sequence also contains a logical interval, whose width is the remaining recorded-value mass. Thus a draw can have small local variance value but large terminal value, or vice versa.

This paper makes four contributions.

1. We first solve the released construction exactly for $N = 2$, yielding a transparent counterexample to global oracle optimality.

2. We prove a sharp arbitrary-N theorem: at fixed $\delta = 1/20$, the repeated oracle's worst-case approximation-ratio supremum is N, the largest any N-round procedure can have.

3. We solve an implementable certified-score problem. For simultaneous AI intervals, descending dollar-weighted interval width is globally optimal, and a covering-knapsack dynamic program is exact for heterogeneous costs.

4. We provide an exact-rational Bellman solver on finite candidate and action grids, independent JSON certificates, exhaustive small-instance checks, and reproducible $N = 1000$ stress tests.

The negative theorem is about the stopping-time objective; it does not contradict Proposition 2. The positive theorem is exact for a robust interval certificate; it does not purport to solve the unrestricted continuous-action betting-CS control problem.

2 Pinned confidence construction

For a candidate total m, the betting process updates

$$W_t(m) = W_{t-1}(m)\{1 + \lambda_t(m)(Z_t - \mu_t(m))\}, \qquad W_0(m) = 1, \tag{2}$$

where

$$Z_t = \frac{\pi_{I_t} f_{I_t}}{q_t(I_t)}, \qquad \mu_t(m) = m - \sum_{s<t} \pi_{I_s} f_{I_s}.$$

Candidates with $W_t(m) \geq 1/\delta$ are rejected. The released ApproxKelly implementation [2] initializes its previous-payoff array at zero, so $\lambda_1 = 0$, and caps $|\lambda_t(m)| \leq 5/2$. We intersect the betting set with its running intersection and the logical interval

$$[L_t, U_t] = \left[\sum_{s\leq t} \pi_{I_s} f_{I_s}, \sum_{s\leq t} \pi_{I_s} f_{I_s} + \sum_{i\in R_{t+1}} \pi_i\right]. \tag{3}$$

The mathematical results use the exact denominator in Z_t. Tiny floating-point stabilizers in the research code are numerical implementation details, not part of the martingale definition. The arbitrary-N witness also lies on a uniform candidate grid allowed by the released code.

Definition 2 of [1] calls q_t a probability distribution on the remaining set, and Proposition 2 optimizes over its simplex. We report literal-simplex results and separately state what happens under strict full support.

3 The exact two-transaction problem

The zero first bet makes the first-round betting set all of $[0,1]$. After sampling item i, the combined interval therefore has diameter $1-\pi_i$. Define

$$A(\pi,\varepsilon)=\{i\in\{1,2\}:1-\pi_i\le\varepsilon\}.$$

Proposition 1 (Complete $N=2$ solution)**.** *For every first action q,*

$$\mathbb{E}_q[\tau]=2-\sum_{i\in A(\pi,\varepsilon)}q(i). \tag{4}$$

Hence the literal-simplex optimum is 2 if $\varepsilon<\min\{\pi_1,\pi_2\}$, and 1 otherwise. If $A\neq\varnothing$, the minimizers are exactly the distributions with $q(A)=1$.

Proof. The first draw stops exactly on A. Otherwise the sole remaining item is audited at time two and (3) is a singleton. Thus $\mathbb{E}_q[\tau]=q(A)+2\{1-q(A)\}$, and optimizing gives the claim. □

For

$$(\pi_1,\pi_2)=\left(\frac{3}{4},\frac{1}{4}\right),\quad (f_1,f_2)=\left(\frac{1}{3},1\right),\quad \varepsilon=\frac{1}{3},\quad \delta=\frac{1}{20},$$

the optimum, prop-M, and oracle expectations are respectively

$$1<\frac{5}{4}<\frac{3}{2}. \tag{5}$$

Under strict full support the infimum remains one but is not attained; putting probability $1-\eta$ on item 1 gives expectation $1+\eta$.

4 A sharp arbitrary-N lower bound

The two-item example understates the failure. We now keep the conventional risk limit $\delta=0.05$ fixed and make the gap approach the entire horizon. Fix $N\ge 2$ and $\rho\in(0,1]$. Set

$$\varepsilon=\frac{1}{20N},\qquad a=\frac{\varepsilon}{2},\qquad w=\frac{a}{N-1}, \tag{6}$$

and define one large transaction and $N-1$ small transactions by

$$\pi_0=1-a,\quad f_0=\frac{\rho w}{1-a};\qquad \pi_i=w,\quad f_i=1\quad(1\le i<N). \tag{7}$$

The contributions are $\rho w, w,\ldots,w$, and all are positive.

Lemma 1 (Two surviving witnesses)**.** *Until transaction 0 is audited, the combined running confidence set contains*

$$m_-=a,\qquad m_+=5a, \tag{8}$$

whose separation is 2ε.

Proof. If k small transactions have been audited, the logical lower endpoint is $kw \le a$; its upper endpoint is at least $\pi_0 > 5a$. Thus both witnesses survive (3).

Under (1), write C_R for the remaining contribution and L for the observed contribution. For every possible next item,

$$Z_t = \frac{\pi_i f_i}{\pi_i f_i / C_R} = C_R, \qquad Z_t - \mu_t(m) = m^\star - m, \tag{9}$$

because $C_R = m^\star - L$. Here $m^\star = a + \rho w$, so at either witness $|m^\star - m| < 2\varepsilon$. The released bet cap gives

$$W_t(m_\pm) \le \left(1 + \frac{5}{2}\,2\varepsilon\right)^t \le \left(1 + \frac{1}{4N}\right)^N < e^{1/4} < 2 < 20 = \frac{1}{\delta}. \tag{10}$$

Neither witness is rejected. They are grid points on the uniform grid with $40N+1$ points, at indices 1 and 5. Since they survive at every round, the running intersection retains both. □

Theorem 1 (Sharp worst-case oracle ratio)**.** *For (6)–(7),*

$$V^\star = 1, \qquad \mathbb{E}[\tau_{\mathrm{oracle}}] = 1 + \frac{N-1}{1+\rho}. \tag{11}$$

Consequently, over N-transaction instances, the worst-case approximation-ratio supremum of the repeated oracle is exactly N.

Proof. Before item 0 is sampled, the lemma makes the combined diameter at least $2\varepsilon > \varepsilon$, so stopping is impossible. Once item 0 is sampled, the remaining recorded-value weight is at most $a \le \varepsilon$, so (3) stops the audit. Hence τ_{oracle} is exactly the rank of item 0.

Sequential probability-proportional sampling is the order of independent exponential clocks having the corresponding rates. For each small item i, the probability that its rate-w clock precedes the rate-ρw clock is $1/(1+\rho)$. Linearity of expectation for the rank gives (11).

Auditing item 0 first leaves logical width a, so the zero first bet and (3) stop in one round. Finally every stopping time lies in $\{1, \dots, N\}$, so no approximation ratio can exceed N; (11) approaches N as $\rho \downarrow 0$. □

Every oracle action has strict full support when $\rho > 0$. If competitors are also required to have strict support, one is an unattained infimum: assign first-round mass $1 - \eta$ to item 0, which gives $\mathbb{E}\tau \le 1 + \eta(N-1)$, and let $\eta \downarrow 0$.

Prop-M stops no later than the rank of item 0, whose expectation is

$$1 + \frac{(N-1)w}{\pi_0 + w} = 1 + \frac{a}{\pi_0 + w} < \frac{1}{\pi_0}, \tag{12}$$

which approaches one. Most strikingly, perfect scores $S_i = f_i$ make prop-MS identical to (1), so perfect point prediction can be maximally bad for this stopping objective.

5 Certified AI intervals: an exact minimax policy

The negative theorem says that point predictions of contribution do not encode terminal uncertainty. Suppose instead that a model, prior audit, or calibrated procedure supplies simultaneous bounds

$$\ell_i \le f_i \le u_i \qquad (1 \le i \le N). \tag{13}$$

After auditing a set B, the exact image of this box under the total is

$$C_B^{\text{box}} = \left[\sum_{i\in B} \pi_i f_i + \sum_{i\notin B} \pi_i \ell_i, \sum_{i\in B} \pi_i f_i + \sum_{i\notin B} \pi_i u_i\right]. \tag{14}$$

Both endpoints are attainable because the uncertainty set is a Cartesian product. Define the dollar-weighted uncertainty

$$d_i = \pi_i(u_i - \ell_i), \qquad D(B) = \sum_{i\notin B} d_i. \tag{15}$$

The observation changes the location of (14), but not its width $D(B)$.

Theorem 2 (Pathwise minimax box policy)**.** *Let $d_{(1)} \geq \cdots \geq d_{(N)}$, and define*

$$k^\star = \min\left\{k \in \{0,\ldots,N\} : \sum_{r=k+1}^{N} d_{(r)} \leq \varepsilon\right\}. \tag{16}$$

Auditing in nonincreasing order of d_i stops (14) in exactly $k^\star$ reviews. Moreover, over all adaptive randomized policies,

$$\inf_q \sup_{f\in\prod_i[\ell_i,u_i]} \mathbb{E}_f[\tau_{\text{box}}(q)] = k^\star. \tag{17}$$

The lower bound holds on every sample path.

Proof. The sorted policy leaves the residual in (16). Any history of length $t < k^\star$ can remove at most $\sum_{r=1}^{t} d_{(r)}$. Its residual width therefore exceeds ε, regardless of observations, randomization, or adaptation. No policy can stop earlier on any path, while the sorted construction attains the bound. □

This gives a precise answer to the implementable robust-certificate problem: audit uncertainty, not merely predicted error. For a multiplicative guarantee, write the condition without division as

$$(1-a)f_i \leq S_i \leq (1+a)f_i, \qquad 0 \leq a < 1. \tag{18}$$

It implies

$$\ell_i = \frac{S_i}{1+a}, \qquad u_i = \min\left\{1, \frac{S_i}{1-a}\right\}. \tag{19}$$

If no upper endpoint clips, then

$$d_i = \pi_i S_i \frac{2a}{1-a^2}. \tag{20}$$

Thus $\pi_i S_i$ is the correct priority, but it should be used as a descending deterministic priority for this certificate, not as a randomized probability. When (19) clips, the exact priority in (15) can differ even in rank from $\pi_i S_i$.

5.1 Heterogeneous costs

If item i costs $c_i > 0$ to review, the exact robust problem becomes

$$\begin{aligned} \min_{x \in \{0,1\}^N} \quad & \sum_i c_i x_i \\ \text{s.t.} \quad & \sum_i d_i x_i \geq D(\varnothing) - \varepsilon. \end{aligned} \tag{21}$$

The repository implements an exact rational Pareto-frontier dynamic program. After each item, a state records removed width and cost, caps width at the target, and discards a state if another has at least as much removed width at no greater cost. Induction over items proves that every discarded partial solution is dominated and that the least-cost terminal state solves (21).

5.2 Risk accounting and a hybrid audit

Suppose (13) holds simultaneously with probability at least $1 - \delta_{\mathrm{AI}}$. On that event, (14) covers $m^\star$ at every time, under any adaptive policy. If a betting CS has simultaneous failure probability δ_{CS}, their running intersection fails with probability at most

$$\delta_{\mathrm{AI}} + \delta_{\mathrm{CS}} \tag{22}$$

by a union bound; independence is unnecessary. A deployable architecture can therefore individually examine a certainty stratum chosen by (15), then run a randomized risk-limiting procedure on the residual population, explicitly allocating the two risk budgets.

This structure is consistent with the distinction in PCAOB AS 2315 between items individually examined and the residual population subject to sampling, while representative residual samples must give population items an opportunity for selection [6]. This observation is operational context, not a legal or professional-compliance conclusion.

6 Finite-grid Bellman verification

For general N, the state must contain the ordered history, remaining set, wealth at every candidate m, ApproxKelly payoff accumulators, and current running intersection. If $\Phi(x, q, i)$ is the transition, the exact equation is

$$V(x) = \begin{cases} 0, & x \text{ is in the stopping region,} \\ 1 + \inf\limits_q \sum\limits_{i \in R(x)} q(i) V(\Phi(x, q, i)), & \text{otherwise.} \end{cases} \tag{23}$$

The action changes both transition probability and confidence state. This is the continuation value absent from the one-step surrogate.

We implement (23) with exact rational arithmetic on a finite candidate grid and the strict-support mesh $q_i \in \{1/D, \ldots, D/D\}$. Backward induction over the finite horizon gives the global optimum over that finite action set; no simulation enters these values. Table 1 uses a 21-point candidate grid and $D = 6$. Policies not lying on this action mesh can beat its optimum, so the table is verification and structure discovery, not a continuous-action optimality claim.

The first mesh action in the mixed $N = 4$ case is $(1/6, 1/6, 1/2, 1/6)$, concentrating on neither the largest recorded value nor the largest contribution alone. This is numerical evidence for a blended continuation-value index, not yet a theorem.

Table 1: Exact expected stopping times on the stated finite grid/mesh.

Case	Oracle	prop-M	prop-MS	Certified priority	Mesh Bellman
Terminal value, $N = 3$	2.4167	2.1500	2.3158	2.0556	2.0556
Mixed information, $N = 4$	3.8558	3.4488	3.8302	3.7176	3.4213
Clipped score, $N = 4$	3.2008	3.4488	3.1839	3.4213	3.4213

Table 2: Synthetic certified-box workload; not a client-data claim.

Scenario	Sorted optimum	prop-MS	Oracle	prop-M	Uniform
Unclipped, concentrated	347	452.13	452.68	493.41	895.66
Clipped, high risk	412	536.52	531.01	528.11	898.89
Diffuse values	740	819.31	819.41	895.66	900.44
Tight certificate	659	736.39	736.64	770.81	975.57

7 Reproducible stress tests

We generated four deterministic $N = 1000$ synthetic populations from fixed seeds. Each AI interval follows (19); the stopping rule is solely the simultaneous-box width. The sorted optimum is exact. Comparator means use 1,000 independent exponential-race permutations, which exactly generate their probability-proportional-without-replacement orders. Table 2 reports mean reviews; Monte Carlo standard errors for prop-MS range from 0.17 to 0.30 reviews.

The exact policy uses 9.7–23.3% fewer reviews than randomized prop-MS in these scenarios. These are controlled stress tests of the theorem's mechanism, not estimates of savings in any audit population. Real validation requires authorized data, simultaneous calibration checks, review-cost measurement, and comparison under the applicable audit objective.

8 Related work and novelty boundary

Waudby-Smith and Ramdas [3] developed confidence sequences for sampling without replacement; Shekhar et al. [1] extended this direction to adaptively weighted monetary totals and AI side information. Imberg et al. [4] study machine-learning-assisted active sampling for finite-population inference. Their optimal importance probabilities use predicted means and covariances to minimize expected asymptotic MSE. That work reinforces the importance of predictive uncertainty, but it does not optimize an RLFA stopping time or the exact robust certificate (14).

Kato and Nakagawa [5] study sequential audit sampling with exact finite-population boundary conditions and expected stopping times. Their model is binary deviation-rate auditing under a hypergeometric random order; they explicitly place monetary-unit sampling outside its direct scope. It neither resolves adaptive weighted selection nor the oracle question here.

Two ten-angle literature searches, including exact-title citation checks and 2024–2026 work, found no prior statement of Theorem 1 or Theorem 2. This is not a novelty guarantee. Before submission, the result should be checked in subscription indexes, sent to the original authors, and independently reviewed by sequential-analysis and audit-methodology experts.

9 Discussion

The sharp lower bound identifies the control-theoretic failure precisely: zero next-step importance variance does not imply fast entry into a stopping region. In the constructed family, a nearly risk-free but very large recorded value has tiny $\pi_i f_i$, so the oracle postpones the one transaction whose review ends the audit.

Certified intervals turn that diagnosis into an exact implementable policy. They expose how much each unaudited transaction can still move the total. Under independent coordinate bounds this uncertainty is modular, which makes descending $\pi_i(u_i - \ell_i)$ pathwise optimal. The result is deliberately conditional on simultaneous calibration. A point model with good average accuracy but no joint coverage guarantee cannot be inserted into (14) while preserving audit risk.

The main open mathematical problem is the continuous-action solution or a sharp approximation guarantee for (23), ideally with correlated calibrated uncertainty rather than boxes. The main empirical problem is to test the hybrid certainty-stratum design on authorized transaction populations without confusing synthetic efficiency with real audit evidence.

Reproducibility

The repository [7] contains exact rational certificates for (5), (11), and the robust example; independent verifiers that do not import the package; exhaustive interval-grid tests; the augmented-state solver; synthetic benchmark generators; and the manuscript source. The sharp certificate's $N = 100, \rho = 1/1000$ instance has

$$\mathbb{E}\tau_{\text{oracle}} = \frac{9091}{91} \approx 99.9011, \qquad \mathbb{E}\tau_{\text{prop-M}} \leq \frac{396001}{395902} \approx 1.00025.$$

References

[1] S. Shekhar, Z. Xu, Z. Lipton, P. Liang, and A. Ramdas. Risk-limiting financial audits via weighted sampling without replacement. In *Proceedings of UAI*, PMLR 216:1932–1941, 2023. https://proceedings.mlr.press/v216/shekhar23a.html.

[2] S. Shekhar et al. *WeightedWoRConfSeq*, commit a834e459a47f. https://github.com/sshekhar17/WeightedWoRConfSeq.

[3] I. Waudby-Smith and A. Ramdas. Confidence sequences for sampling without replacement. *Advances in Neural Information Processing Systems*, 33, 2020. https://arxiv.org/abs/2006.04347.

[4] H. Imberg, X. Yang, C. Flannagan, and J. Bärgman. Active sampling: A machine-learning-assisted framework for finite population inference with optimal subsamples. *Technometrics*, 67(1):46–57, 2025; online 2024. https://doi.org/10.1080/00401706.2024.2374554.

[5] M. Kato and K. Nakagawa. Sequential audit sampling with statistical guarantees, arXiv:2604.06116, 2026.

[6] Public Company Accounting Oversight Board. AS 2315: Audit Sampling (effective December 15, 2026). https://pcaobus.org/oversight/standards/auditing-standards/details/as-2315--audit-sampling-(effective-on-12-15-2026).

[7] R. Sater. *RLFA Optimal Policy: exact certificates and certified-score sampling.* https://github.com/RichieSater/rlfa-optimal-policy, 2026.